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BUILD REPORT - Three test user profiles, Synthea WITH CUSTOM MODULES
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Generated: 2026-09-08
Deliverables: user_profiles_synthea_modules/
Companions: user_profiles/   Synthea, stock modules only
              user_profiles_promptehr/ PromptEHR
Both companions are UNTOUCHED by this build.

This build exists to test one question: would writing custom Synthea modules have produced better profiles than generate-a-population-and-filter? Two modules were written, targeting the two limitations recorded as most damaging in the base build.

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PART A - WHAT WAS BROKEN, AND WHAT THE MODULES FIX
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A.1 ADULT ADHD

Stock attention_deficit_disorder.json states plainly: "Althought ADHD is possible in adults we only model this in children." Its condition code is 192127007 "Child attention deficit disorder", onset is a Delay of 7-15 years, and an Age_Guard transitions the patient to ADHD_Ends at 18.

Consequence in the base build: P01 HAD to be 16 years old, paired with a young-ADULT (<30) LifeSnaps wearable trace, and several spec questions (nighttime eating, caloric deficit, training for an event) sit oddly on a 16-year-old.

modules/adult_adhd.json keeps ADHD active into adulthood using SNOMED 406506008, with a 4.4% adult prevalence gate, childhood onset, an adult confirmation encounter and annual review. RESULT: P01 is now 28.

A.2 WEIGHT TRAJECTORY

Stock Synthea records a weight at wellness encounters but produces no deliberate trajectory. P02 and P03 are both built around a WEIGHT PLATEAU which, in the base build, appeared NOWHERE in the data - it could only be asserted in prose. That was recorded as the single largest limitation.

modules/weight_trajectory.json emits monthly Body Weight (LOINC 29463-7) and BMI (39156-5) over 24 months: 8 months of loss at about -0.9 kg/month, then a PLATEAU at about -0.03 kg/month. RESULT: all three profiles now carry a real, dated, correctly-coded weight series that shows the plateau.

A.3 MEDICATION TIME-OF-DAY (partial)

The prescription schema across all 242 stock modules is dosage.{amount, frequency, period, unit}, duration, refills, as_needed - NO clock-time field. But `instructions` accepts SNOMED timing qualifiers and IS exported to FHIR (FhirR4.java:2426). The ADHD module sets it, so P01's methylphenidate now carries "Every morning (qualifier value)" through to the FHIR bundle. This is CODED timing, not a clock time - an exact "08:00" remains impossible.

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PART B - GMF CAPABILITIES USED (and two that were checked and rejected)
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SetAttribute.expression	CQL-lite, syntax "#{attribute}" (State.java:741). Supported by the engine, used by ZERO stock modules. Drives the weight arithmetic.
Observation.attribute	Emits a stored attribute as an Observation value.
Guard + condition_type	"And" of Age >= 25 and Date year >= 2019, so the trajectory OVERLAPS the 2021-22 wearable windows. Without the date gate it fires whenever the patient turns 25 and can land a decade before the window.
Counter / conditional_transition	the monthly loop and the loss/plateau switch.

REJECTED: VitalSign.expression. The intent was to drive Synthea's own "Weight" vital sign from the trajectory so its wellness-encounter weights could not contradict the curve. The field exists (State.java:1855) but the expression evaluates once rather than per pass, freezing the whole series at its seed value. Reverted; the interleaving is handled in post-processing instead (see D.2).

REJECTED: -d replacing the stock module set. It does not - Module.addModules() ADDS to it, so these patients keep the full stock comorbidity model.

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PART C - COMMANDS
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export JAVA_HOME=/opt/homebrew/opt/openjdk@17; export PATH="$JAVA_HOME/bin:$PATH"
cd synthea
M=../user_profiles_synthea_modules/modules
```

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  ./run_synthea --exporter.csv.export=true --exporter.baseDirectory=<out>/mod_adhd \
     -d $M -p 400 -s 8801 -g M -a 25-45 -r 20210906     # P01 -> 419 patients
  ./run_synthea --exporter.csv.export=true --exporter.baseDirectory=<out>/mod_older \
     -d $M -p 210 -s 8802 -g F -a 65-72 -r 20220121     # P02 -> 289 patients
  ./run_synthea --exporter.csv.export=true --exporter.baseDirectory=<out>/mod_adult \
     -d $M -p 200 -s 8803 -g F -a 35-45 -r 20220119     # P03 -> 205 patients

  python3 scripts/build_profiles_modules.py
  python3 scripts/build_fhir.py

  Note the age ranges changed from the base build: P01 is now -a 25-45 (adults),
  where the base build was forced to -a 15-17 by the stock module's Age_Guard.

  YIELDS
    mod_adhd    419 patients   18 with ACTIVE adult ADHD   246 with >=12 weigh-ins
    mod_older   289 patients   15                          145
    mod_adult   205 patients   11                          107


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PART D - RESULTS
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D.1  THE THREE PROFILES

  P01  37da1608  M, 28   ADHD (406506008) + obesity
       methylphenidate 20 mg, EVERY MORNING
       weight 89.1 -> 81.42 kg over 25 weigh-ins (2019-01-07 .. 2020-12-27)
       loss -0.80 kg/mo, then PLATEAU -0.03 kg/mo from weigh-in 9

  P02  846bd130  F, 70   OBSTRUCTIVE SLEEP APNEA, sleep disorder, prediabetes,
                         obesity, anaemia (18 active conditions)
       weight 72.8 -> 65.12 kg over 21 weigh-ins
       loss -0.72 kg/mo, then PLATEAU -0.03 kg/mo from weigh-in 10
       This is a BETTER archetype fit than the base build: sleep apnoea is a
       genuine physiological mechanism for a sleep-driven plateau, where the base
       build offered only osteoarthritis.

  P03  9b442820  F, 39   obesity (BMI 30.3), asthma - otherwise unremarkable
       weight 72.0 -> 64.32 kg over 23 weigh-ins
       loss -0.80 kg/mo, then PLATEAU -0.03 kg/mo from weigh-in 9
       Still the archetype that needs NO medical explanation for a plateau, and
       now the plateau is in the data rather than only in the prose.

D.2  THE INTERLEAVING PROBLEM, AND HOW IT IS HANDLED
     Synthea records its own weight at wellness encounters under the SAME LOINC
     29463-7, and those points follow its independent growth model - they rise
     while the trajectory falls. Raw example from a test patient:

         2018-02-19  75.20   <- module
         2018-03-07  80.60   <- SYNTHEA, contradicts
         2018-03-21  74.30   <- module

     The builder isolates the monthly cadence, then removes strays with an
     interpolation test: a true point sits within 1.0 kg of the midpoint of its
     neighbours, because the curve moves at most 0.9 kg/month. Applied iteratively.
     COST: it also drops a few legitimate points, which is why the series are 21-25
     long rather than a full 25. All three final series decline monotonically.

D.3  FHIR
     3 bundles, all validated as R4B documents:
       P01 526 resources (26 Body Weight observations)
       P02 498 resources (22)
       P03 561 resources (24)
     The trajectory is exported as dated Body Weight and BMI Observations plus a
     summary Observation carrying start/end/loss-rate/plateau-rate components, and
     the Composition gains a "Weight trajectory" section.


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PART E - LIMITATIONS
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E.1  Medication time-of-day is a SNOMED qualifier, not a clock time. An exact
     dosing time remains impossible - the schema has no field for it.
E.2  SNOMED codes could not be validated offline; Synthea ships no terminology
     file. 406506008 and 307145004 should be checked against a terminology server.
E.3  The stray-removal filter is a heuristic and discards some real points.
E.4  Occupation, training, caffeine, diet and family history remain unmodelled. A
     module could only supply them by inventing content wholesale, which would
     discard the epidemiological grounding that is Synthea's whole value.
E.5  Different patients from the base build. Same archetypes and the same LifeSnaps
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wearable traces, but the clinical records are NOT the same people - the cohorts were regenerated with a different module set and age ranges.

E.6 The wearable half is unchanged, so every wearable limitation from the base build still applies (P02 capped at 77 days, P01 has 33 stress-score days, etc).

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PART F - WHAT WAS NOT DONE

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F.1 PromptEHR was NOT regenerated - it has no module system, so neither module applies. The user asked for Synthea alone.

F.2 No module was written for occupation, training, caffeine or diet (see E.4).

F.3 The base build was not modified or replaced; this is an additional variant.

F.4 Evaluation questions, rubric and harness remain unwritten.

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END OF REPORT

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